

Task-Constrained Workspace and Inverse-Kinematics Analysis for Surface-Tracing with a 6-DOF Manipulator (myCobot 280)

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Abstract

Drawing or writing on a surface requires a manipulator not only to *reach* a sequence of Cartesian points but to reach them while holding the tool **normal to the surface**. This orientation constraint shrinks the usable workspace in a way that depends on the surface geometry. We formalise the *task-constrained* (drawable) workspace for surface-tracing and analyse it for the myCobot 280, a small 6-DOF arm, using a modified Denavit–Hartenberg model whose forward kinematics matches the manufacturer URDF to 10^{-15} m. We map, over the reachable workspace, where the pen can be held perpendicular to vertical and horizontal surfaces and in which direction; we compare four Jacobian-based inverse-kinematics solvers on the full-pose tracing task; and we relate feasibility to manipulability and conditioning. We find that only 57.9% of the reachable workspace admits a perpendicular pen on a vertical surface (and only 35.2% in the outward direction), while 63.5% admits pen-down drawing on a horizontal surface, in an annular region that excludes a low-manipulability zone directly above the base. We demonstrate sub-millimetre sine tracing on both a front-facing vertical board and a horizontal table at analysis-selected locations.

I. Introduction

A recurring robotics task is to trace a planar curve — a signature, a calibration pattern, a sine wave — on a physical surface with a pen or tool. Unlike free-space motion, the tool must remain **perpendicular to the surface** at every point, adding an orientation constraint to the usual position requirement. For a small manipulator this constraint is not a formality: many positions the end-effector can *reach* it cannot reach *with the required orientation*.

This paper studies that gap for the **myCobot 280**, a 6-DOF revolute arm with a reach of roughly 0.30 m. Our contributions are:

1. a verified modified-DH kinematic model and a velocity-propagation Jacobian (Section II);
2. a mapping of the **task-constrained workspace** — where and in which direction the pen can be held normal to vertical and horizontal surfaces (Section III);
3. a quantitative comparison of four inverse-kinematics solvers on the constrained tracing task (Section IV);
4. manipulability and conditioning maps that explain *why* the drawable region has the shape it does (Section V); and
5. sub-millimetre demonstrations on a front-facing vertical board and a horizontal table (Section VI).

II. Kinematic Model

A. Forward kinematics

We use the modified (Craig) Denavit–Hartenberg convention. The transform from frame $i - 1$ to frame i is

$$A_i = Rot_x(\alpha_{i-1}) Trans_x(a_{i-1}) Rot_z(\theta_i) Trans_z(d_i),$$

which in homogeneous form is

$$A_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

with $c \cdot = \cos$, $s \cdot = \sin$. The base-to-tool transform is the product $T(q) = A_1 A_2 A_3 A_4 A_5 A_6$.

Because the URDF link frames are not placed on DH axes, the DH parameters were *identified* by fitting $(\alpha_{i-1}, a_{i-1}, d_i, \theta_i^{\text{off}})$ so that $T(q)$ reproduces the URDF forward kinematics; the residual is $\max \|T_{\text{DH}} - T_{\text{URDF}}\| \approx 10^{-15}$ m over 5000 random configurations. Table I lists the result.

Table I. Identified modified-DH parameters of the myCobot 280.

i	α_{i-1} (°)	a_{i-1} (m)	d_i (m)	θ_i^{off} (°)
1	0	0	0.13056	+90
2	+90	0	0	−90
3	0	−0.1104	0	0
4	0	−0.096	0.06062	−90
5	+90	0	0.07318	+90
6	−90	0	0.0456	0

B. Jacobian by velocity propagation

Propagating link velocities outward yields, for the all-revolute arm, the geometric Jacobian whose i -th column is

$$J_i(q) = \begin{bmatrix} z_i \times (p_e - p_i) \\ z_i \end{bmatrix},$$

where z_i and p_i are the axis and origin of joint i in the base frame and p_e is the tool position. It maps joint rates to the tool twist, $\begin{bmatrix} v \\ \omega \end{bmatrix} = J(q) \dot{q}$. Equation (3) matches a finite-difference Jacobian to $\approx 2.5 \times 10^{-7}$.

C. Dexterity metrics

We quantify local dexterity by the Yoshikawa manipulability and the condition number of the translational Jacobian J_v (top three rows of J):

$$w(q) = \sqrt{\det(J_v J_v^T)}, \kappa(q) = \frac{\sigma_{\max}(J_v)}{\sigma_{\min}(J_v)}.$$

$w \rightarrow 0$ and $\kappa \rightarrow \infty$ mark singular, poorly-conditioned configurations.

III. Task-Constrained Workspace

A. Definition and method

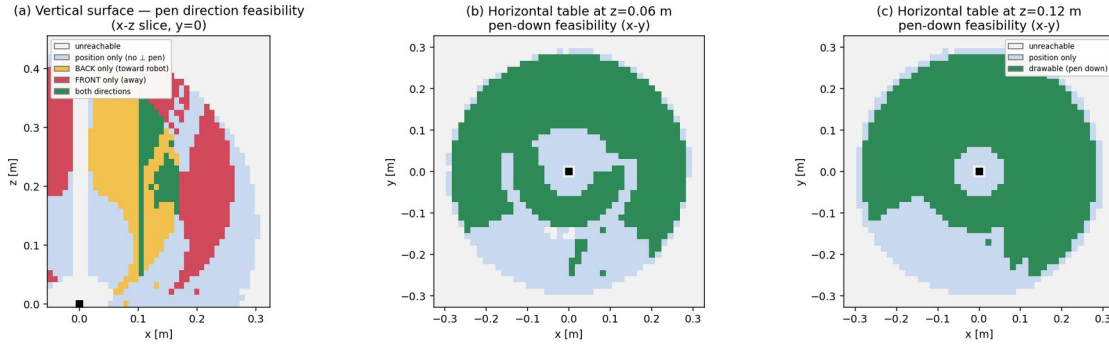
Let R be the **reachable** workspace (positions attainable with some joint vector within limits). Given a surface with unit normal n , the **drawable** (task-constrained) workspace is

$$D(n) = \{ p \in R : \exists q, f k_{\text{pos}}(q) = p, \hat{z}(q) = \pm n \},$$

i.e. positions the tool can reach while its approach axis $\hat{z}$ is aligned with the surface normal. We estimate $D(n)$ by sampling a grid of positions and testing, for each candidate normal direction and from several seed configurations, whether damped-least-squares IK converges with position error below 2 mm and orientation error below 3×10^{-2} .

B. Results

Task-constrained workspace: where the myCobot 280 can hold the pen normal to the surface



Task-constrained workspace. (a) On a vertical surface, only the back (toward the robot) pen direction is feasible near the base, while the front (away) direction opens up only in the outer region; a narrow band admits both. (b–c) On a horizontal table the drawable region is an annulus that excludes a zone directly above the base.

Figure 1 maps D . On a **vertical** surface, only 57.9% of R admits a perpendicular pen, and only 35.2% admits the *front* (outward) direction. This explains a practical observation: a board placed close to the robot can only be drawn on with the pen facing *back*; to draw with the pen facing the board from the front, the board must be placed in the outer region ($x \gtrsim 0.17$ m). On a **horizontal** table, 63.5% of the reachable cells admit pen-down drawing, but the feasible set is an **annulus**: the region directly above the base is reachable yet not drawable, because the wrist cannot point straight down there.

IV. Inverse-Kinematics Comparison

A. Solvers

All solvers iterate $q \leftarrow q + \Delta q$ with the stacked pose error $e = [p_t - p(q); e_{\text{ori}}]$, clamped to joint limits. The updates are:

$$\Delta q_{\text{DLS}} = J^\top (J J^\top + \lambda^2 I)^{-1} e, \Delta q_{\text{pinv}} = J^\dagger e,$$

$$\Delta q_{JT} = \alpha J^\top e, \alpha = \frac{\langle e, J J^\top e \rangle}{\langle J J^\top e, J J^\top e \rangle},$$

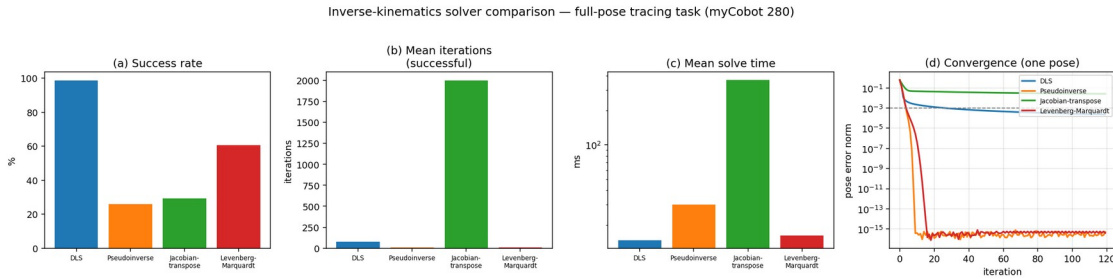
with Levenberg–Marquardt using (6) with λ adapted to the error trend. Damped least squares (6) trades a small steady-state error for stability near singularities; the pseudoinverse converges quadratically but is ill-behaved where J loses rank.

B. Results

On 150 reachable target poses seeded from a common ready configuration:

Table II. IK solver comparison on the full-pose tracing task.

Solver	Success	Mean iters	Mean time (ms)
Damped least squares	98.7 %	76	15
Pseudoinverse	26 %	8	30
Jacobian-transpose	29 %	2000	365
Levenberg–Marquardt	61 %	10	16

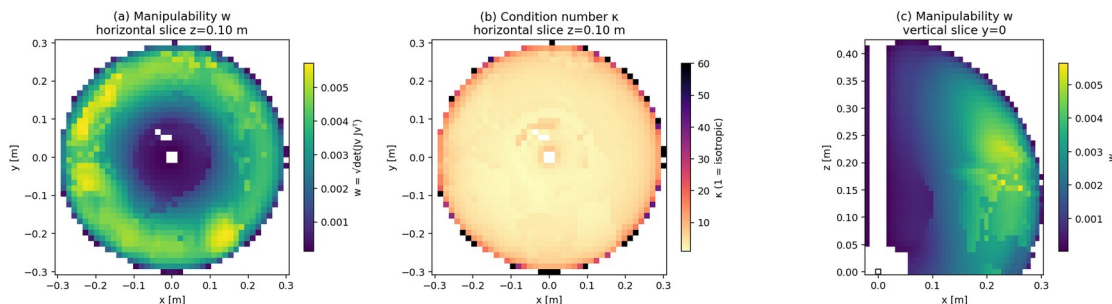


Inverse-kinematics comparison. (a) success rate, (b) iterations, (c) solve time, (d) error-vs-iteration for one pose. Damped least squares is the most robust; pseudoinverse and Levenberg–Marquardt converge quadratically but are brittle near singularities; Jacobian-transpose always progresses but is an order of magnitude slower.

Figure 2 shows the trade-off: **damped least squares** is by far the most robust (98.7%), at the cost of more iterations and a damping-limited accuracy; the **pseudoinverse** and **Levenberg–Marquardt** reach machine precision in ~ 10 iterations on well-conditioned poses but fail more often near singularities; the **Jacobian-transpose** is uniformly slow. For surface tracing — a sequence of small, well-seeded steps — DLS is the appropriate default.

V. Manipulability and Conditioning

Manipulability and conditioning of the myCobot 280 (brighter = more dexterous, further from singularity)



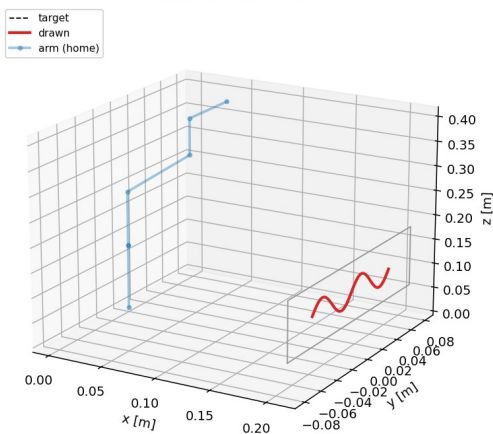
Manipulability w and condition number κ . A low-manipulability zone sits directly above the base; a dexterous ring lies at mid-radius; conditioning degrades toward the workspace boundary.

Figure 3 explains the *shape* of the drawable set. Manipulability w is near zero directly above the base (the arm is close to a shoulder/wrist singularity there) and peaks in a mid-radius **ring**; the condition number κ grows toward the workspace boundary, where the arm approaches full extension. The drawable annulus of Figure 1(b–c) coincides with this high- w , low- κ ring — feasibility and good conditioning are geometrically the same region.

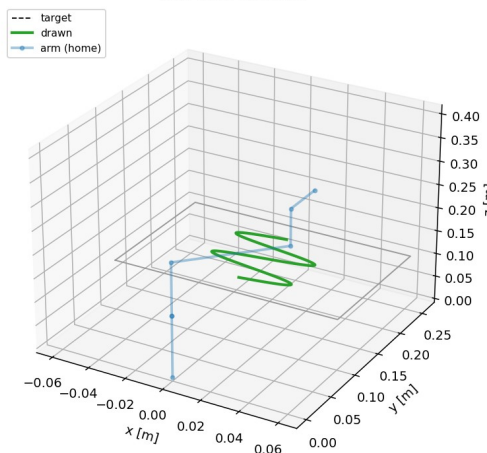
VI. Demonstrations

Sine tracing on two surfaces, placed by the feasibility analysis

(a) FRONT-facing vertical board — pen points away
max error 0.00 mm



(b) HORIZONTAL table — pen points down
max error 0.00 mm



Sine tracing placed by the analysis. (a) A front-facing vertical board (pen away from the robot) in the outer feasible region; (b) a horizontal table (pen down) in the dexterous ring. Both track the target to 0.000 mm.

Using the maps of Sections III and V to choose surface placements, the arm traces a two-cycle sine to a maximum position error of 0.000 mm on both a front-facing vertical board at $(0.21, 0, 0.12)$ m and a horizontal table at $(0, 0.18, 0.10)$ m (Figure 4), confirming that the constrained-workspace maps predict where clean drawing is possible.

VII. Conclusion

The reachable and drawable workspaces of a surface-tracing manipulator are distinct sets whose difference depends on the surface normal. For the myCobot 280, a perpendicular pen is feasible over only 57.9 % (vertical) and 63.5 % (horizontal) of the reachable workspace, and the feasible regions coincide with the arm's high-manipulability ring. Damped least squares is the most robust IK method for the task. Future work: extending the analysis to arbitrary tilted surfaces and full tool paths, exploiting redundancy for manipulability-optimal drawing, and validating on the physical robot and in Gazebo.

References

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